

Predicting ED-to-Ward Waiting Times in Singapore Public Hospitals

Group 6_P2

Colab Notebook: <https://colab.research.google.com/drive/1lqcWt1dMSLVg4hnLa7oh26gVzNsYyxhR?usp=sharing>

Github: <https://github.com/Corv123/Machine-Learning-Proj>



Member	Role
Goh Jing Wen (2403444)	Pipeline, Ablation, Failure Analysis
Corvan Chua (2402844)	Feature Engineering (Lag, Rolling)
Ang Xuan Yu Pamela (2401870)	Visualisations, Outlier Research
Hayden Chua Shao En (2402425)	Alternative Models, Disease Integration
Jeanie Cherie Chua Yue-Ning (2403263)	Business Analysis, Safety Margins

Executive Summary & Stage 1 Recap

Stage 1 Baseline

Model	MAE	RMSE	R ²
Linear Regression	2.439310	3.666226	0.401722

Stage 2 Result

Model	MAE	RMSE	R ²
Random Forest	1.403	2.441	0.742

42.5% MAE Reduction ✓



Key Improvements:

- sklearn Pipeline (prevents data leakage)
- Temporal features (Lag_1_WaitTime)
- External Data integration (Disease 2023-2026)
- Hyperparameter tuning via ablation

Stage 2 Data Engineering

New Features Engineered

Temporal Features

- Lag_1_WaitTime: Previous day's waiting time
- Lag_3_WaitTime_Avg: 3-day rolling average
- Rolling_3d_Attendance: Recent attendance trend
- BOR_Change: Daily occupancy change

Date	north	south	east	west	central	PSI_Max	PSI_Avg
1/4/2014	60	58	59	67	59	67	60.6

External Data

- Disease data (2023 - 2026): Dengue, HFMD, ARI, Diarrhoea
- 209 weeks, 100% coverage of hospital data
- + 1.8% MAE improvement

Date	Day	Hospital	Waiting Ti	Attendanc	BOR	Is_Holiday
1/1/2023	Sunday	AH	1.6	64	80.5	1
1/1/2023	Sunday	CGH	7.7	351	89.6	1
1/1/2023	Sunday	KTPH	2.9	286	96.4	1
1/1/2023	Sunday	NTFGH	7.9	252	91.1	1
1/1/2023	Sunday	NUH(A)	2.1	257	78.3	1
1/1/2023	Sunday	SGH	1.3	309	79.3	1
1/1/2023	Sunday	SKH	3	333	86	1
1/1/2023	Sunday	TTSH	3.8	336	91.2	1

Epi_Week	Date_Range	Week_Start	Week_End	dis_Dengu	dis_HFMD	dis_ARI	dis_Diarrhoea
1	1/1/2023 - 7/1/2023	1/1/2023	7/1/2023	279	9.555556	2559.556	352
2	8/1/2023 - 14/1/2023	8/1/2023	14/1/2023	278	14.36364	2400.182	375.0909091
3	15/1/2023 - 21/1/2023	15/1/2023	21/1/2023	272	15.27273	2404.545	382.3636364

Tested but not included

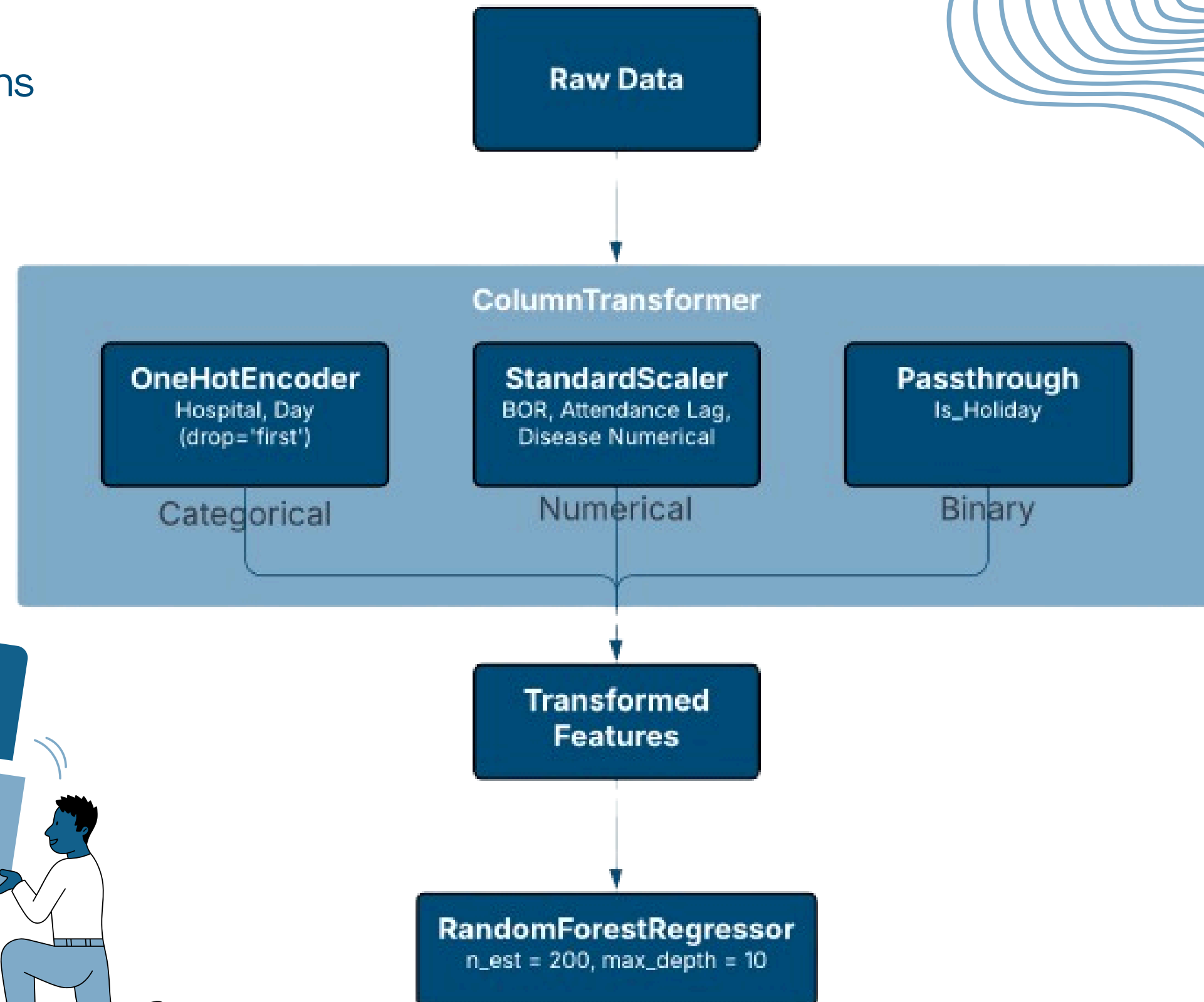
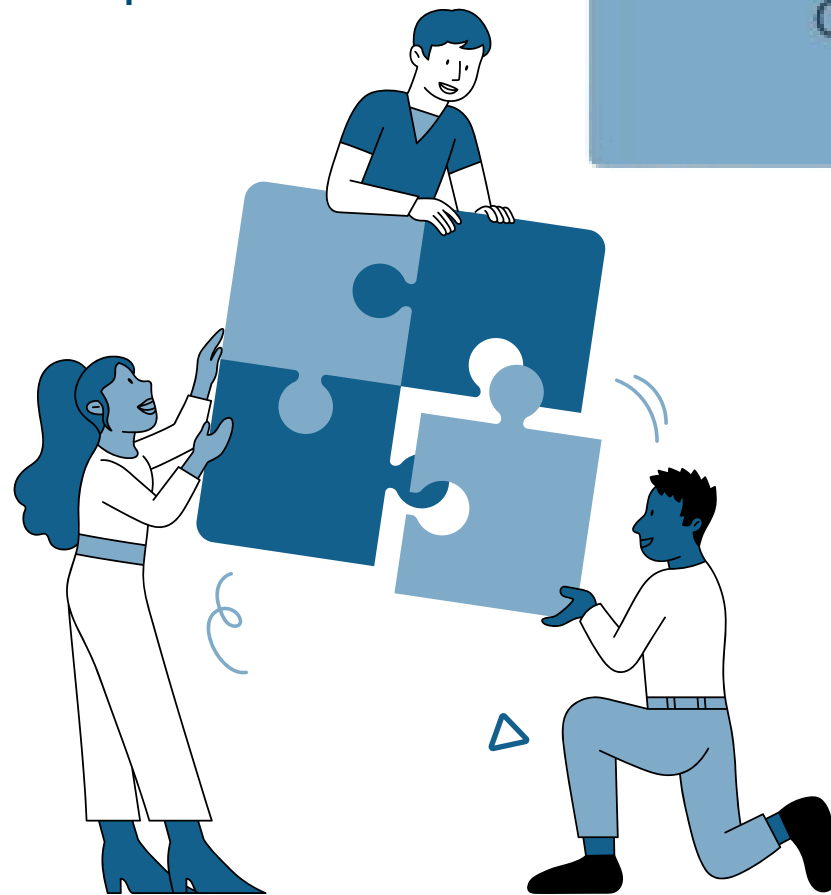
- PSI/Haze data: No significant improvement

Pipeline Architecture

Raw data → Column Transformer → Model → Predictions

Why this prevents data leakage?

- Scaler fitted ONLY on training fold
- No test data information leaks into preprocessing
- Cross-validation applies full pipeline per fold



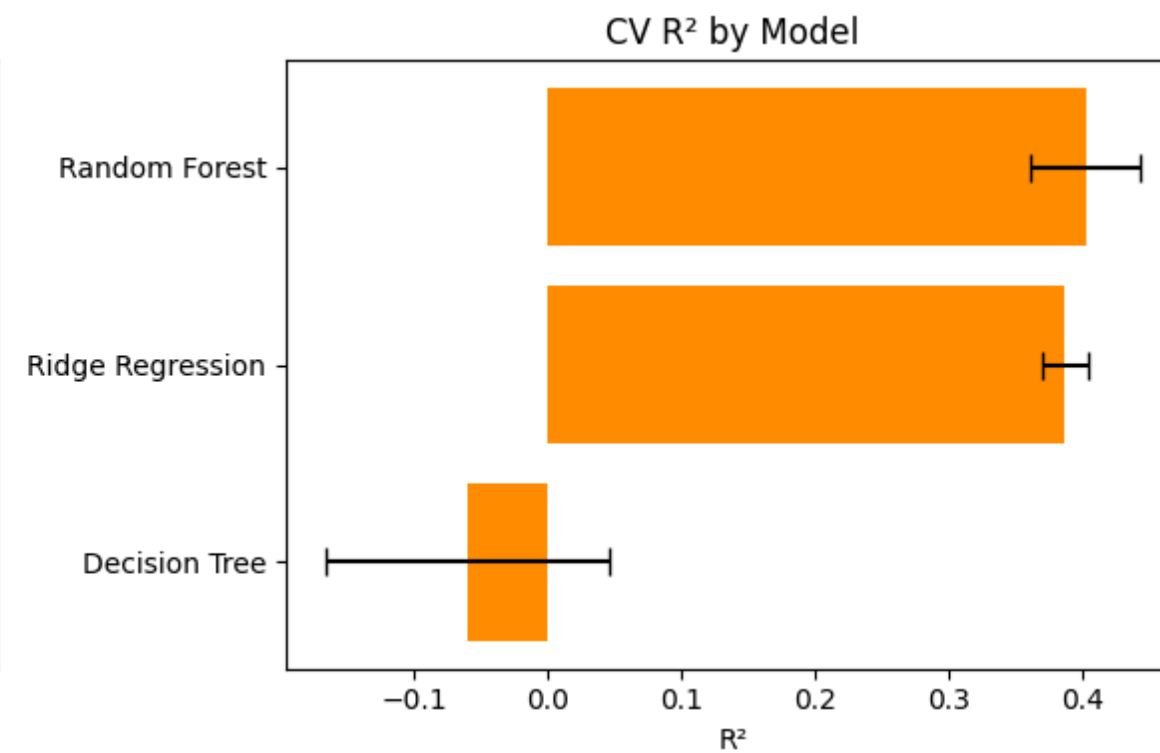
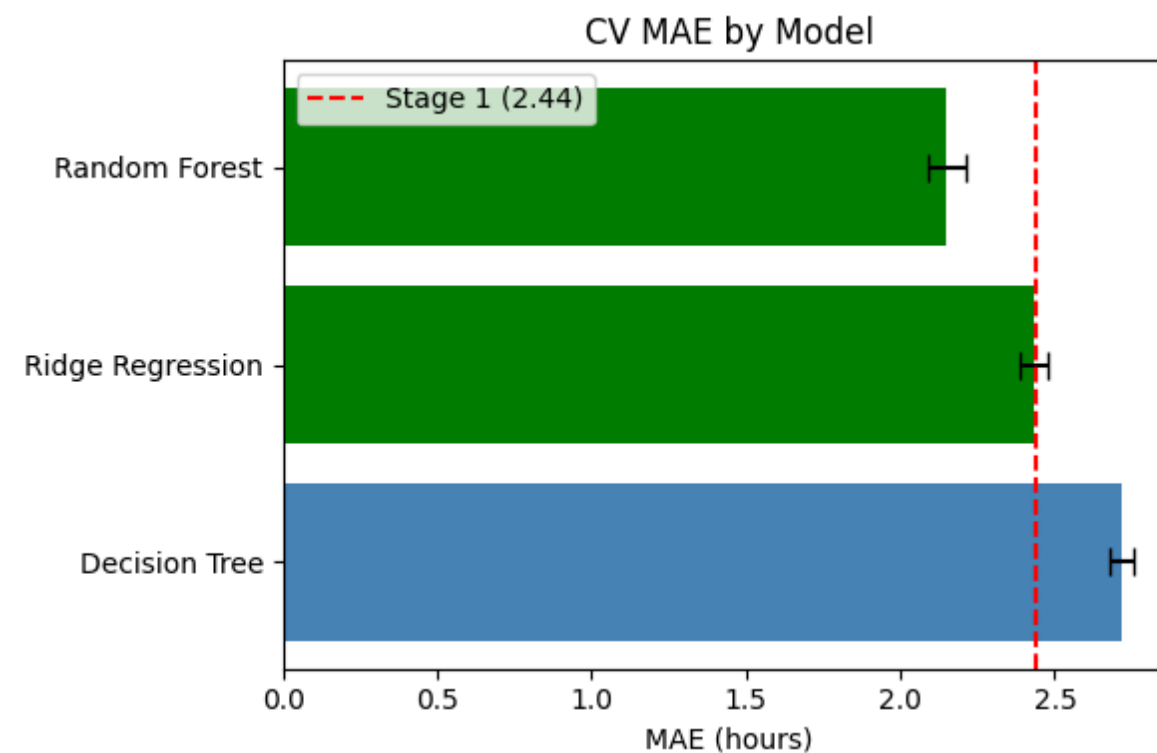
Model Selection

Same setup: sklearn Pipeline · target = wait (hours) · 5-fold CV (shuffled)

Family 1 — Linear (regularized): Ridge → MAE ~2.43 · $R^2 \sim 0.39$

Family 2 — Single tree: Decision Tree → MAE ~2.72 · $R^2 \sim -0.06$

Family 3 — Ensemble (bagging): Random Forest → MAE ~2.15 · $R^2 \sim 0.40$ → best CV MAE (vs Stage 1 LR 2.44 h)



Justification of the “Champion” Model

- **Champion (Sec 6): Random Forest** — set in code after reviewing Sec 3 & 5
- **Why RF:** Best of Ridge / Tree / RF on same CV; below 2.44 h on average MAE (~2.15 h)
- Tuned with ablations: 200 trees, max_depth 10, min_samples_leaf 10, lags on
- Train-period CV MAE (after tuning): ~1.47 h
- Held-out test (Sec 8): MAE ~1.40 h, $R^2 \sim 0.74$ — ~42.5% better than 2.44 h Stage 1

Controlled Ablation & Tuning Log

Controlled Ablation Experiements

Exp	Hypothesis	Values Tested	Best
1	More trees reduce variance	[50, 100, 150, 200]	200
2	Depth limit reduces overfitting	[None, 5, 10, 15, 20]	10
3	Lag captures temporal patterns	[True, False]	True
4	Larger leaves reduce overfitting	[1, 5, 10, 20]	10

Featured Set Comparison

Feature Set	MAE	R ²
Baseline	2.308	0.473
+ Lag Features	1.656	0.716
+ Disease Features	2.028	0.608
+ Lag + Disease	1.661	0.719

Final Model Stability Metrics

Final Model Configuration

- Random Forest (n_est = 200, max_depth = 10, min_samples_leaf = 10)
- Features: Lag + Disease

Cross Validation Stability (5-Fold)

Metric	Mean	Std	Interpretation
MAE	1.427	0.039	Stable across folds
R ²	0.742	0.020	Consistent performance

Final Test Results (Held-out 20%)

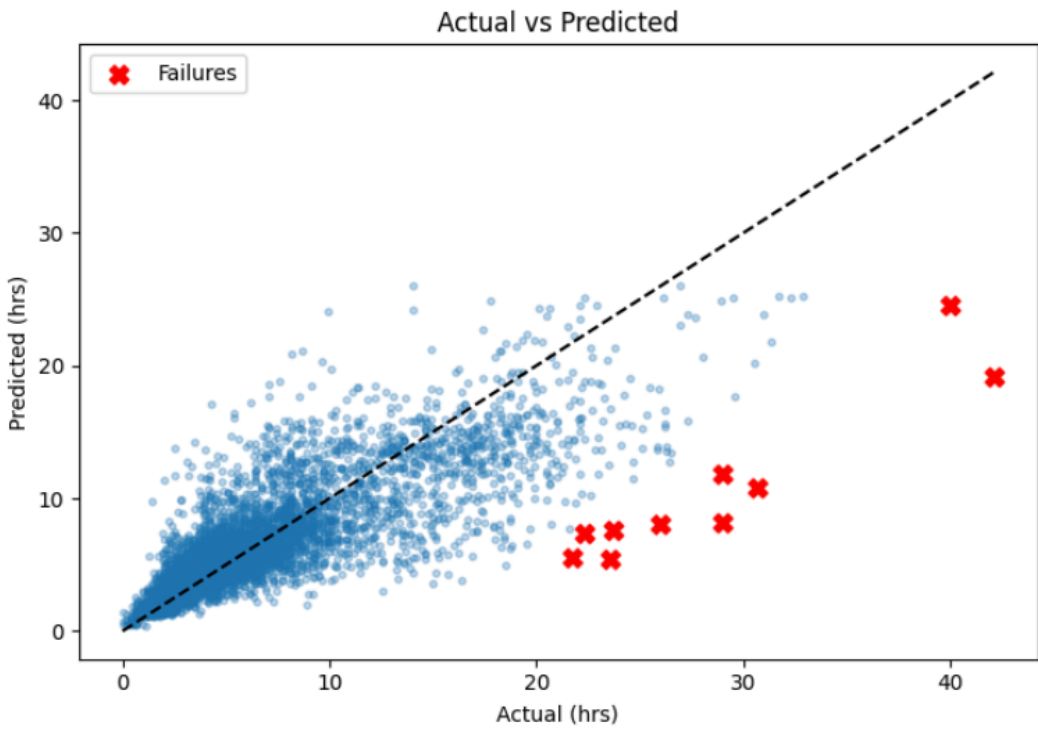
Metric	Value
MAE	1.403 hrs
RMSE	2.441 hrs
R ²	0.742

Mechanical Failure Analysis

TOP 10 WORST PREDICTIONS

Hospital	Day	Actual	Predicted	Error
KTPH	Monday	42.1	19.17	22.93
KTPH	Monday	29.0	8.17	20.83
CGH	Sunday	30.7	10.79	19.91
CGH	Wednesday	23.6	5.43	18.17
SKH	Friday	26.0	8.08	17.92
CGH	Thursday	29.0	11.81	17.19
KTPH	Thursday	21.7	5.54	16.16
KTPH	Tuesday	23.7	7.62	16.08
KTPH	Monday	40.0	24.54	15.46
CGH	Monday	22.3	7.36	14.94

ACTUAL VS PREDICTED



FAILURE PATTERNS

- Over-predictions: 0
- Under-predictions: 10
- Average Failure Wait: 28.8 hrs
- Most Common Hospital: KTPH

Proposed Fixes for Observed Failures

FIX 1

FAILURE OBSERVED

Model under-predicts extreme wait times on high-BOR days

PROPOSED FIX

Extend lag window – add Lag_7 and weekly rolling average to better capture sustained surge patterns

FIX 2

FAILURE OBSERVED

Outbreak and public holiday surge days still produce large errors

PROPOSED FIX

Incorporate real-time MOH disease alert levels and public holiday proximity as explicit features

FIX 3

FAILURE OBSERVED

Certain hospitals appear repeatedly in the top 10 worse predictions

PROPOSED FIX

Train KTPH sub-models to account for each hospital's unique operational patterns

Business Decision Reasoning

- **Errors aren't symmetric:** Under-prediction (actual wait > forecast) vs over-prediction (actual < forecast)
- **Under-prediction (higher real-world cost):** patient stress / unexpected long wait; hospital unprepared, surge unmanaged; risk framed as HIGH (missed surge planning, deterioration) in the notebook's framework
- **Over-prediction (lower cost):** shorter wait than expected; hospital slight over-staffing; risk LOW — manageable
- **Evidence from failure analysis:** all top-10 worst errors were under-predictions; example cited: KTPH Monday ~42 h actual vs much lower predicted wait
- **Decision:** Shift published forecast upward, recommended +2.72 h (MAE + 0.5×Std, “Moderate” option vs 1.47 h MAE-only and 3.97 h aggressive in the notebook output)
- **Rule on slide (one line):** Published wait $\approx$ model output + 2.72 h (wording as in notebook: “Model Prediction + margin”)

Individual Contributions

Name/ID	Contribution
Ang Xuan Yu Pamela 2401870	Created 4 EDA visualizations; Built Hospital*Day heatmap;
Goh Jing Wen 2403444	Built sklearn Pipeline; Conducted baseline CV; Ran ablation studies (4 experiments); Performed failure analysis; Integrated team results; Final test evaluation
Hayden Chua Shao En 2402425	Tested Gradient Boosting & Stacking; Tuned GB hyperparams; Integrated disease data (2023-2026); Tested PSI/haze data
Jeanie Cherie Chua Yue-Ning 2403263	Developed error cost framework; Calculated safety margins; Documented external factors; Proposed future improvements
Corvan Chua 2402844	Developed temporal feature engineering (Lag_1, Lag_3_Avg, Rolling_3d_Attendance, BOR_Change); Compared 4 feature sets; Analyzed feature importance



References

References

- “Weekly Infectious Diseases Bulletin 2023.” Communicable Diseases Agency, 2023, www.cda.gov.sg/resources/weekly-infectious-diseases-bulletin-2023/.
- “Weekly Infectious Diseases Bulletin 2024.” Communicable Diseases Agency, 2024, www.cda.gov.sg/resources/weekly-infectious-diseases-bulletin-2024/.
- “Weekly Infectious Diseases Bulletin 2025.” Communicable Diseases Agency, 2025, www.cda.gov.sg/resources/weekly-infectious-diseases-bulletin-2025/.
- “Weekly Infectious Diseases Bulletin 2026.” Communicable Diseases Agency, 2026, www.cda.gov.sg/resources/weekly-infectious-diseases-bulletin-2026/.

Tools Used

- Google Colab

Links

- Colab Notebook: <https://colab.research.google.com/drive/1lqcWt1dMSLVg4hnLa7oh26gVzNsYyxhR?usp=sharing>
- Github: <https://github.com/Corv123/Machine-Learning-Proj>